



The Future of AI Interconnects: Open Approaches to High-Performance AI Infrastructure

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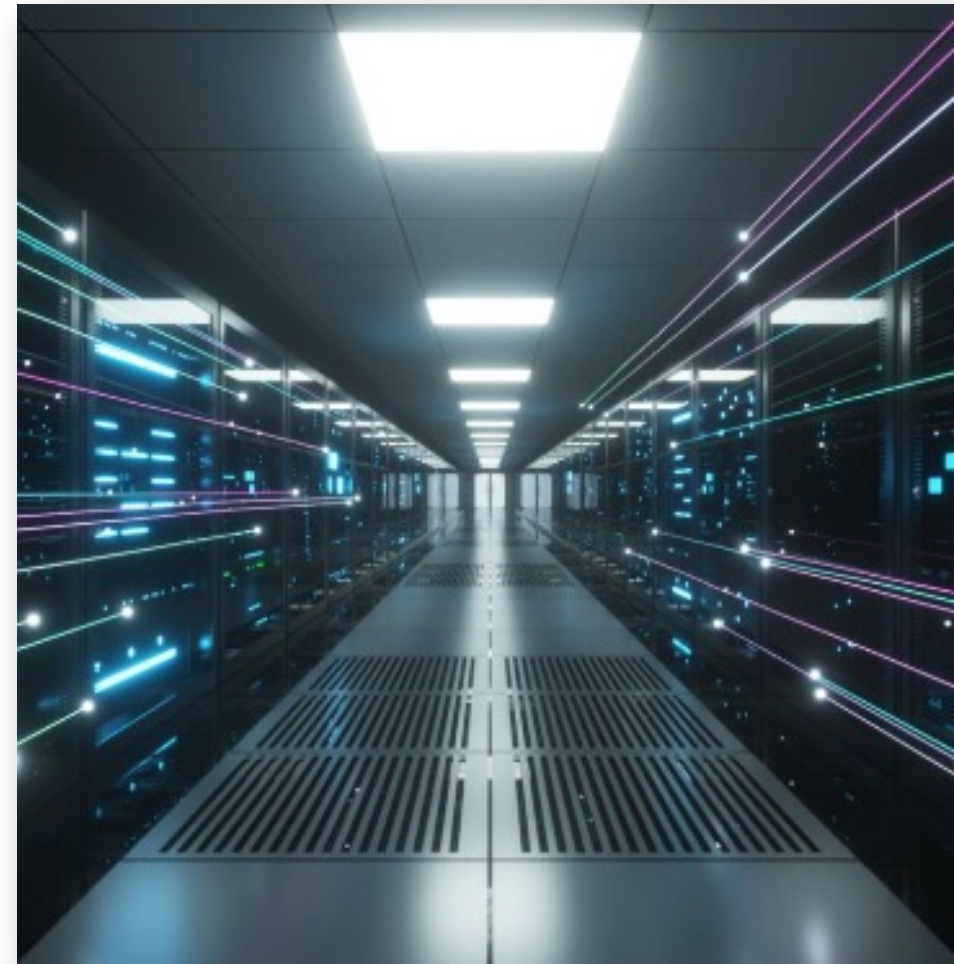
“The network is the computer.”

John Gage, Sun Microsystems, 1984

Four decades on, that idea defines AI: the interconnect now sets the pace of compute.

Outline

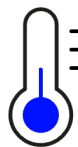
- 1 Why AI interconnects are changing
- 2 The fundamental scaling challenges
- 3 **Go faster:** higher speed electrical interfaces
- 4 **Go wider:** new packaging and optical architectures
- 5 Matching technologies to scale up, scale out, and scale across
- 6 The role of open ecosystems



Remarkable era of innovation



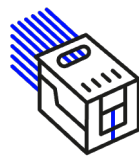
More network bandwidth
is driving acceleration across every layer



Cooling

Old thinking (early 2024)
Having liquid cooling
is just too expensive.

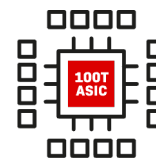
NOW
Not having liquid cooling
is just too expensive.



Host Bus

Old thinking (early 2024)
PCB trace to front panel
is how it is done.

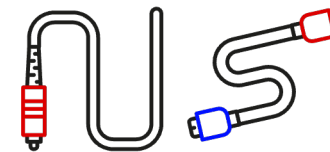
NOW
Co-packaged copper
already here for 224G,
path to 448G underway



Co-packaged optics

Old thinking (early 2025)
CPO challenges.
Maybe sometime in the future.

NOW
CPO ramping this year.

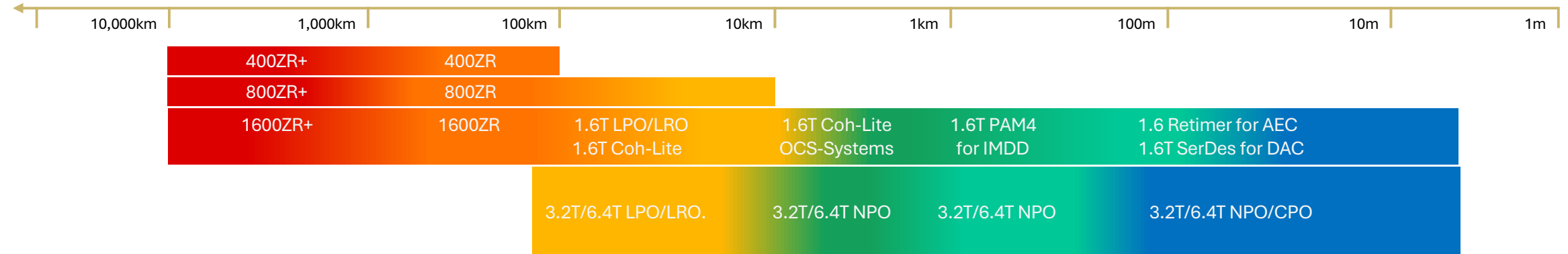


Optical/Copper

Old thinking (Jan 2026)
We'll stay with copper in
scale up for as long as we can.

NOW
Optical scale up
next year.

5



Fundamental Interconnect challenges

Scale Up

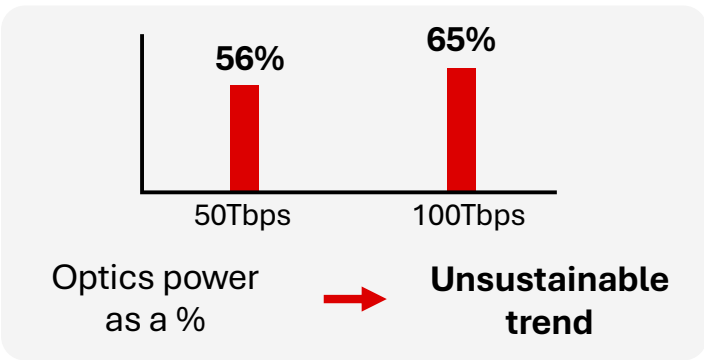
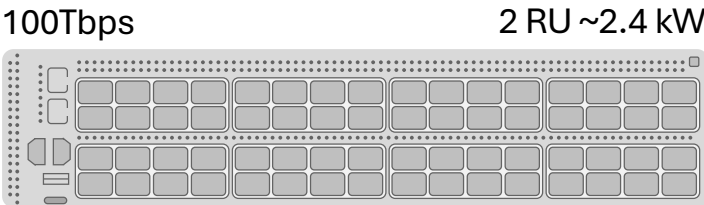
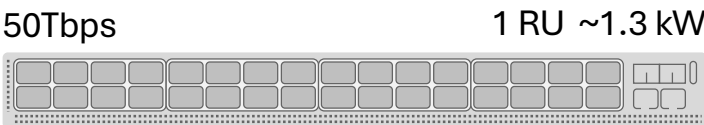
Passive Copper getting shorter



**Passive cables most reliable, lowest cost.
Too short for many applications.**

Scale Out

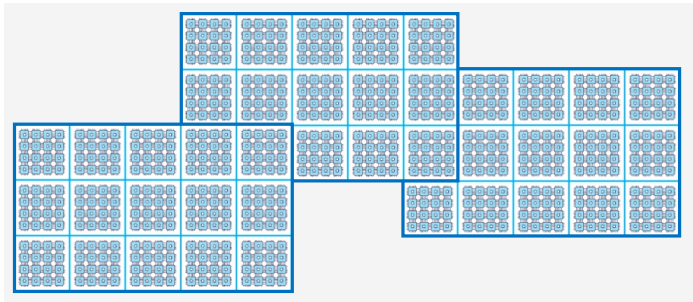
**Insufficient front-panel density
Power scaling issues**



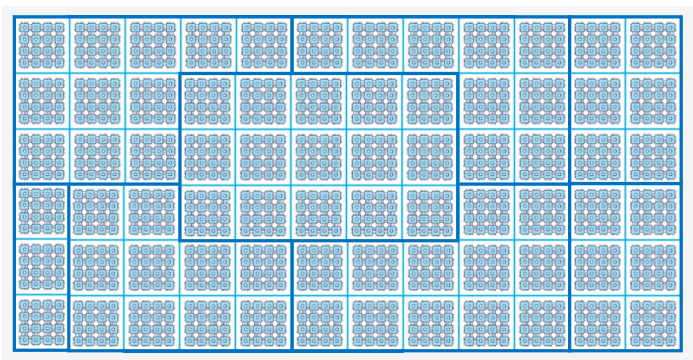
Scale Across

**Lossless connectivity
now required over distance**

LLM training - from 10s of thousands of xPUs



To millions of xPUs



Industry exploring diverse solutions for deployment

Scale Up

Passive Cables getting shorter

Scale Out

**Insufficient front-panel density
Power scaling issues**

Scale Across

**Lossless connectivity
now required over distance**

Individual hyperscalers implement optimized solutions – costs magnified at scale

Active cables

- Active electrical cables
- Active copper cables

Optical replacement

- Co-packaged optics
 - Scale out 2026
 - Scale up 2027/2028
- Optical circuit switch

Linear solutions

- Half-retimed solutions
- Linear pluggable optics
- Co-packaged optics / copper
- Near package optics / copper

Coherent optical

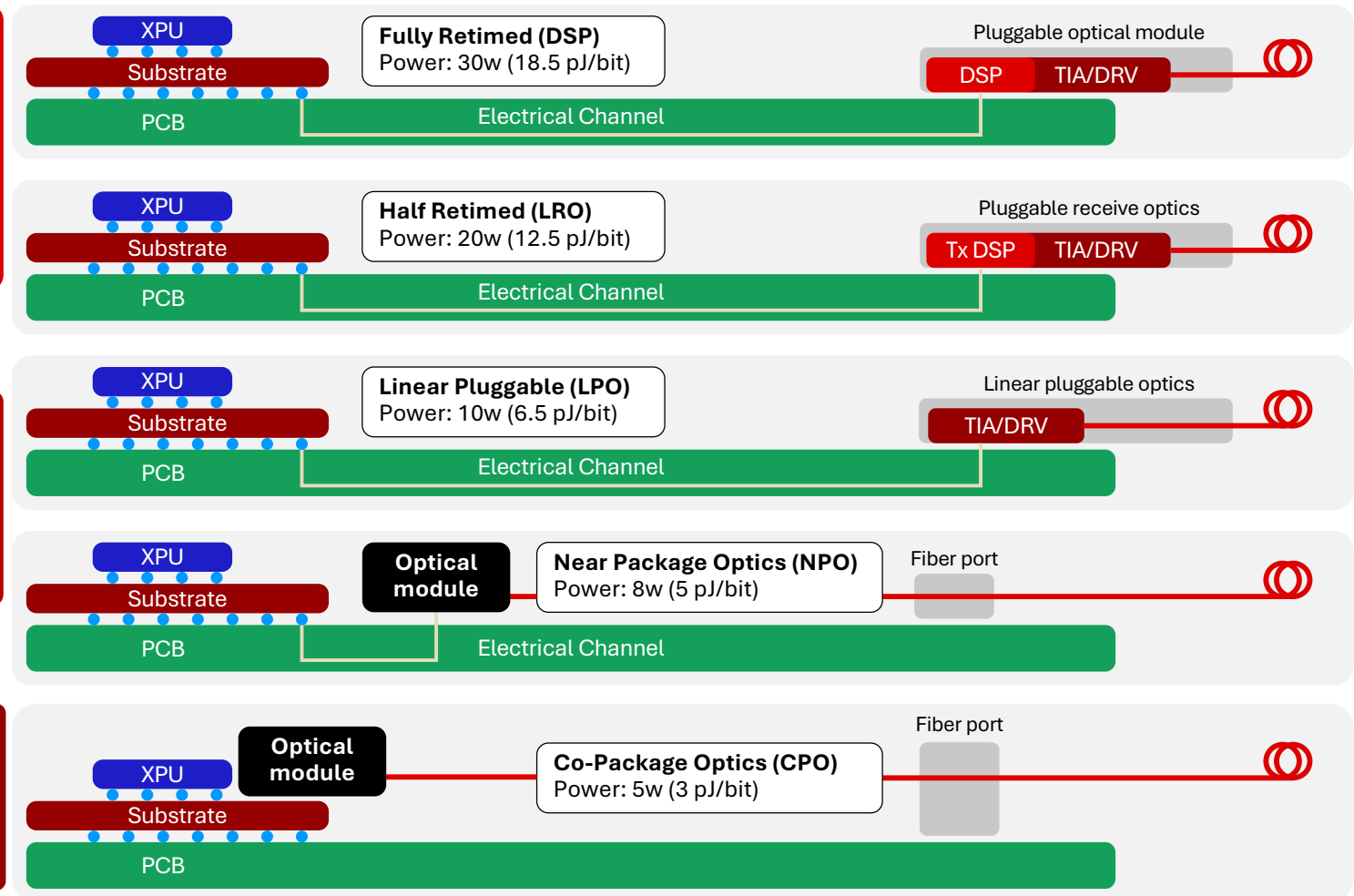
- C&L DWDM systems to maximize connectivity
- Coherent pluggables
 - ZR/ZR+ at 400G, 800G
 - Coherent-Lite, ZR-
- Hyper-rail/multi-rail amplifier configurations

Cracking the XPU beachfront bottleneck with optics

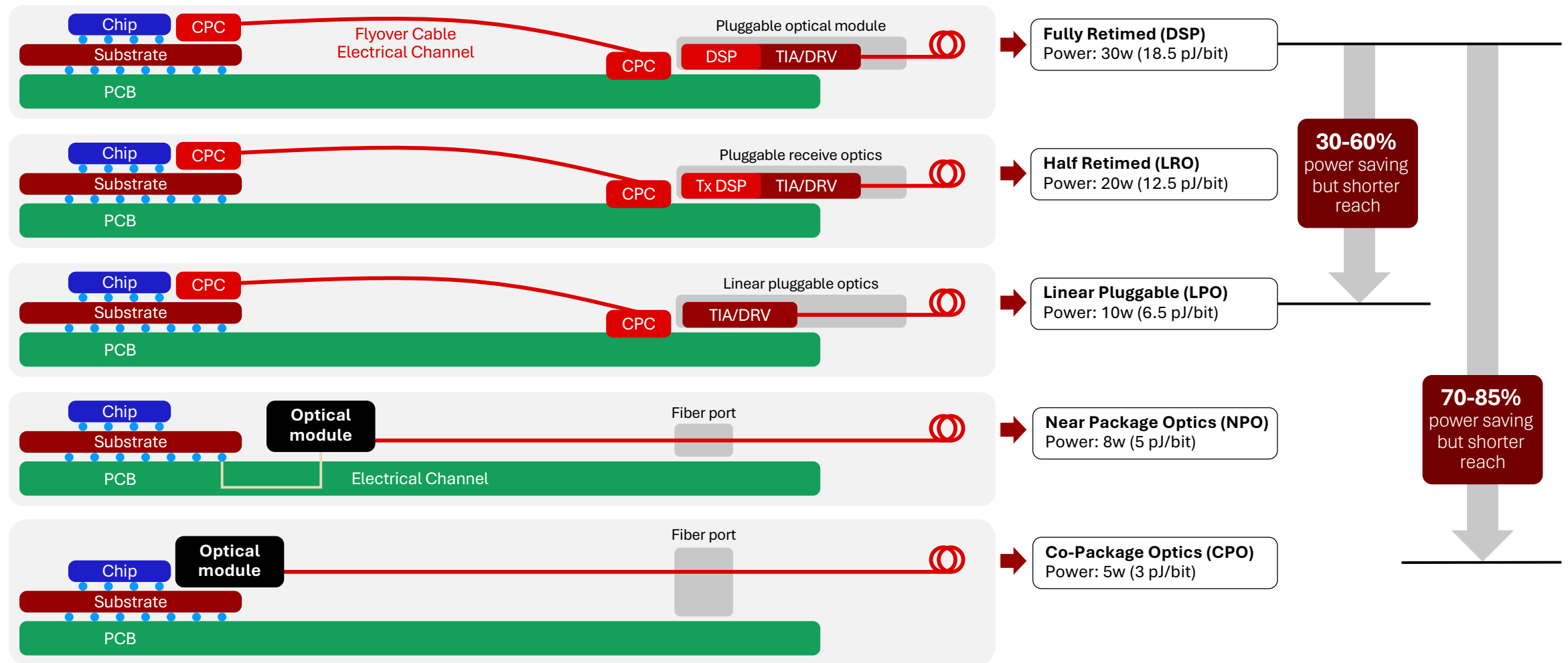
DSP based, retimed links are a fundamental component of data center networks today as they provide robust connectivity among other benefits.

At AI scale, the costs of these benefits are enormous - measured in 100's of MW's and billions of dollars.

Further, at 448Gbps PCB connectivity to front panel breaks down, demanding a new paradigm.



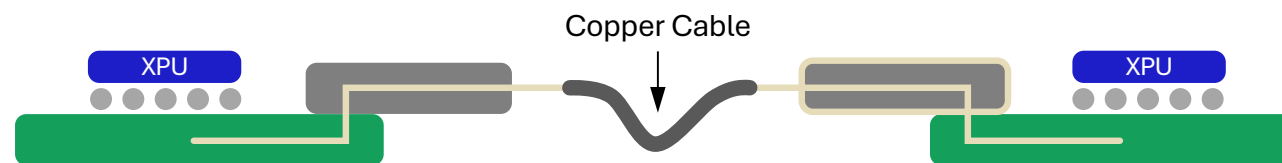
Cracking the XPU beachfront bottleneck with optics using CPC



Shorter electrical paths and less retiming unlock major power savings — driving optics progressively closer to the XPU.

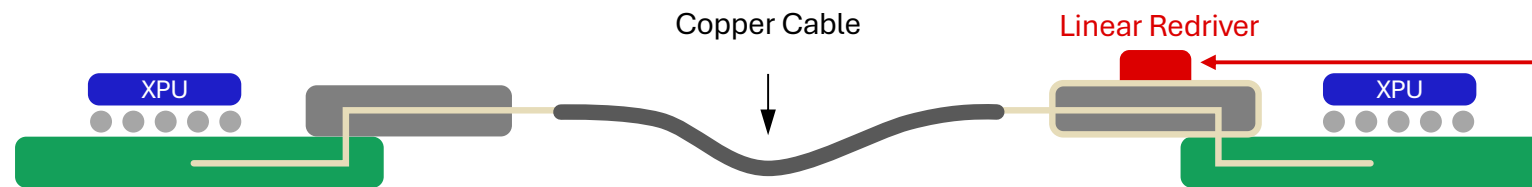
Cracking the XPU beachfront bottleneck with copper

As speeds have increased, and port counts have exploded, it has become financially advantageous to develop alternative forms of connectivity based on copper.



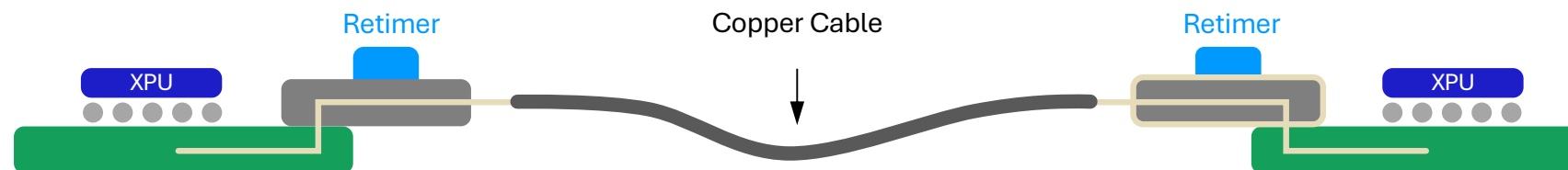
Direct Attach Copper

Zero power
Zero added latency
Max reach: ~1.5m



1.6T Active Copper Cable

1.5pJ/bit / head end
Negligible latency
Max reach: 3-4m



1.6T Active Electrical Cable

10pJ/bit / head end
10's of ns latency
Max reach: 5m

Scale Up in the post-DAC era

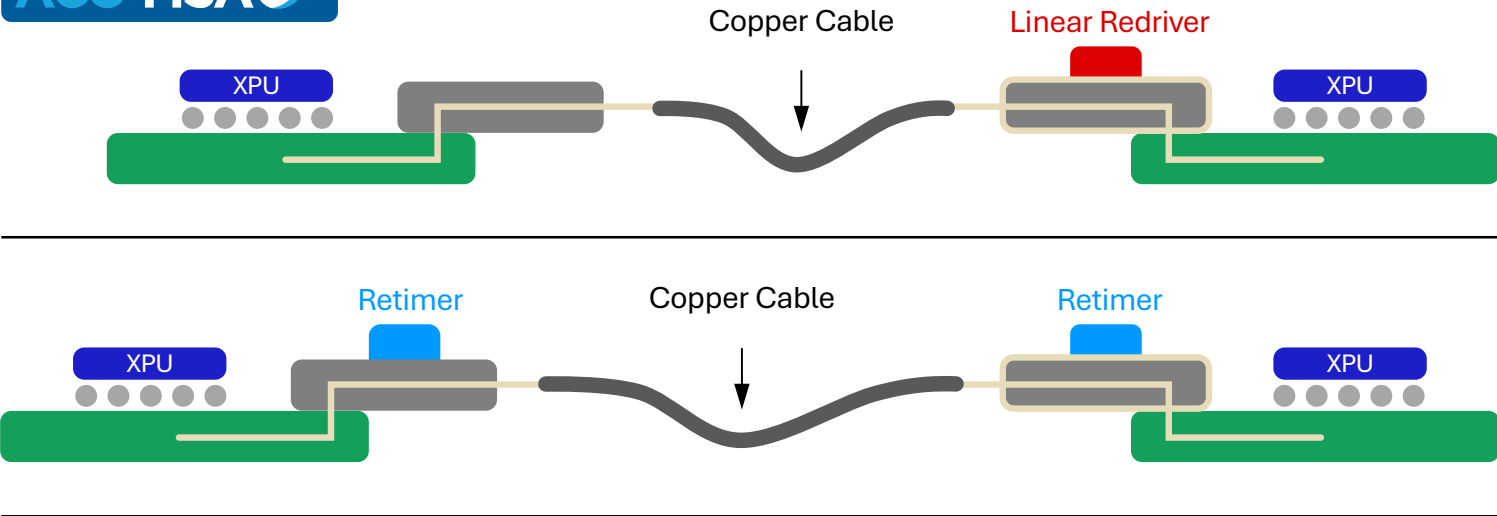
Scale up architects pushing for both intra-rack and rack-to-rack for larger XPU Pods
Longer links required beyond DAC capabilities

8 x 200 Gbps Active Cable Comparison

	AEC	ACC
Length	5m+	Up to 4m
Power per end	10pJ/bit	1.5pJ/bit
Added latency	~35 ns	Zero
Complexity	High	Low
Cost	\$\$\$\$	\$
Technology	Captive	Open MSA

Complexity:

- Configuration & software
- # of power supplies
- Advanced assembly for KGD



ACC extends copper reach while preserving the simplicity, efficiency, and low latency of DAC.

Example implementation: Nitro 2004 linear redriver

Zero-touch ACC

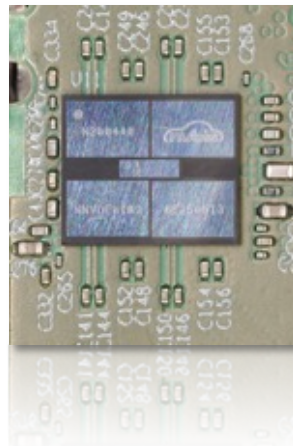
Extends 200Gbps Copper Interconnect

- Cable lengths to 4m
- Intra-rack or rack to rack
- **Up to 70dB channel loss budget**
- **Automatically extends** 1m DAC to 3m



DAC – like Simplicity

- uP free - saves cost & complexity, **faster to deploy**
- No tuning – plug and play
- Mounts on standard PCB
- No power rail design needed, uses only standard 3.3V supply



Efficient Design to Enable Massive Scaling

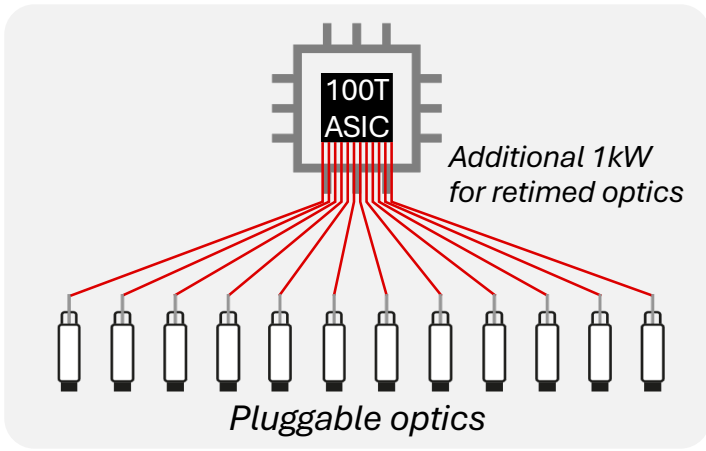
- Rx direction only
- 1.5pJ/bit power consumption
- Zero added latency
- Scales to **448G**



Scale Out evolution

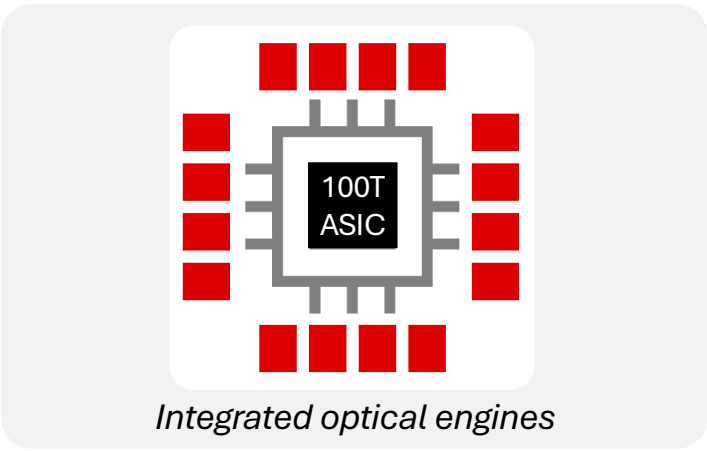
Current

ASIC to Front Panel
Retimed Optics



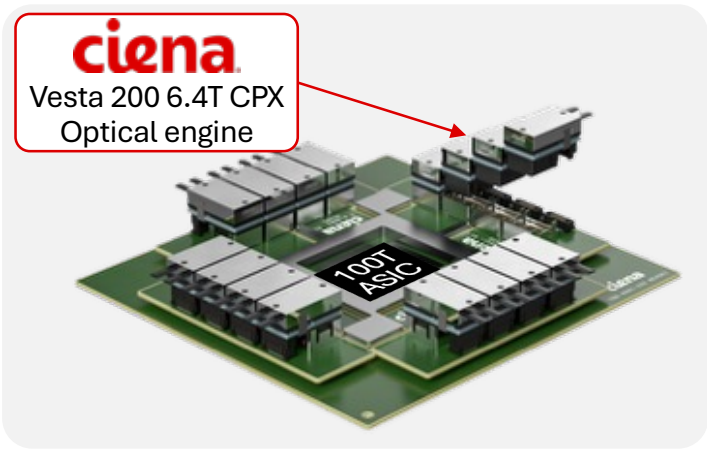
Emerging

Single Vendor
Co-packaged optics

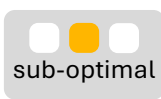


Optimized

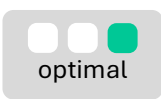
Open Ecosystem
Co-packaged optics



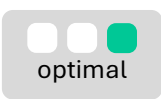
Technology



- Increasingly high cost to retime in dollars and Watts



- Retimers removed
- Power reduced with lower loss host



- Retimers removed
- Power reduced with lower loss host

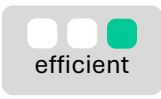
Interconnect supply chain



- Multiple vendors compete
- Deliver 10's of millions annually



- ASIC + interconnect offered as unit
 - Reduced buyer power
 - Reduced supply chain



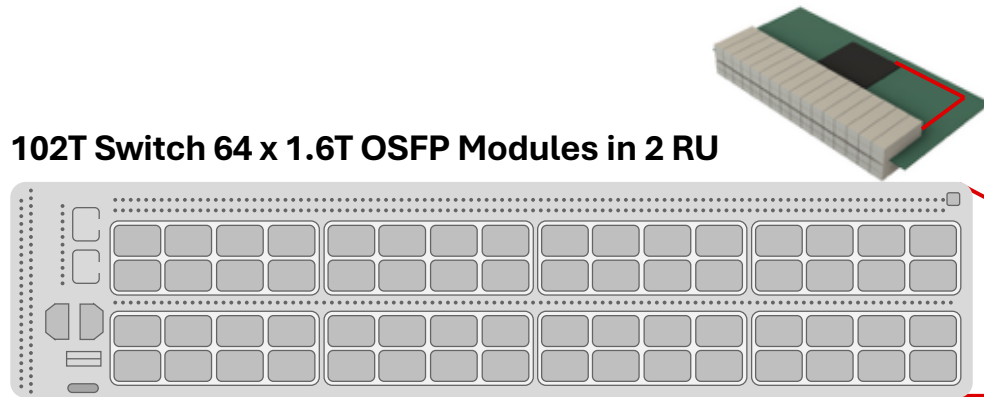
- 6.4T module standardized
 - Open CPX MSA
 - Robust, multi-vendor supply

Increasing density with co-packaging

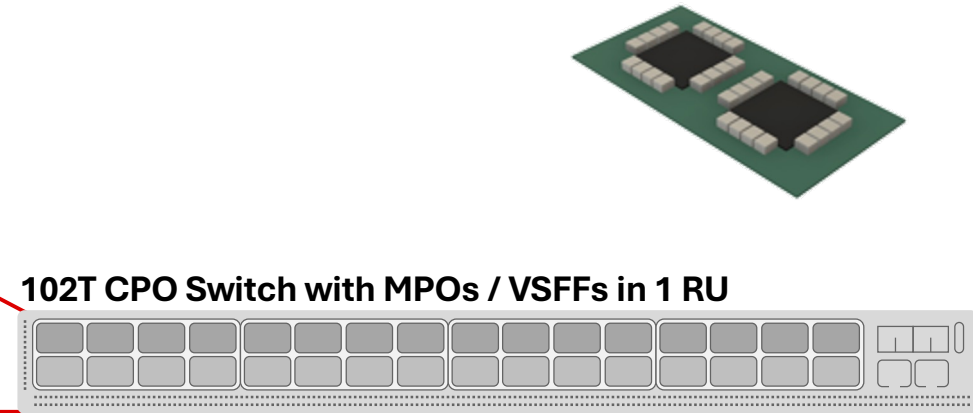
100T Example

Co-packaging increases density versus pluggable architectures

102T Switch 64 x 1.6T OSFP Modules in 2 RU



102T CPO Switch with MPOs / VSFFs in 1 RU



Front-panel pluggables

Long traces → Retimers → **MORE POWER**

Faceplate area = Pluggable area
1RU = 32 OSFP = 256 lanes = 50T @200G

Number of ASICs per tray limited by faceplate

Systems spread out → **HIGHER-POWER LINKS**

Co-packaged I/O

Short traces → Retimer-free → **LESS POWER**

Faceplate area = Connector area
1RU = 250 MMC connectors = 2000 lanes = 400T @200G

Number of ASICs per tray limited by liquid cooling

Denser systems → **LOWER-POWER LINKS**

MSAs ensuring robust multi-vendor ecosystems



Announced on Feb 23, 2026



Announced on Mar 12, 2026



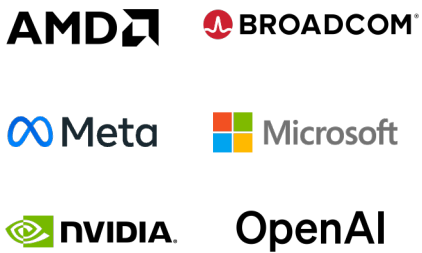
OPEN CPX MSA

Announced on Mar 12, 2026



OCI-MSA

Announced on Mar 12, 2026



High-speed SerDes

As long as systems also use copper



YES

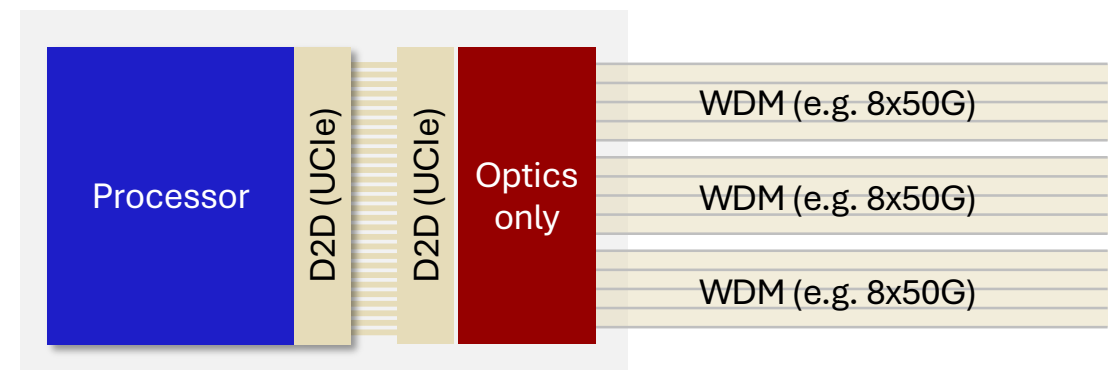
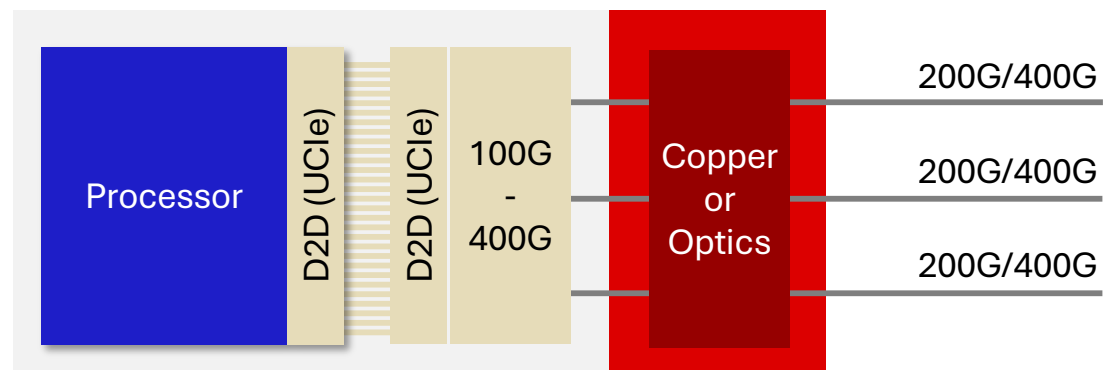
Copper
used?

NO



**Use high-speed SerDes
with Copper or Optics**

**Use Chiplet integration
with Wide & Slow Optics**

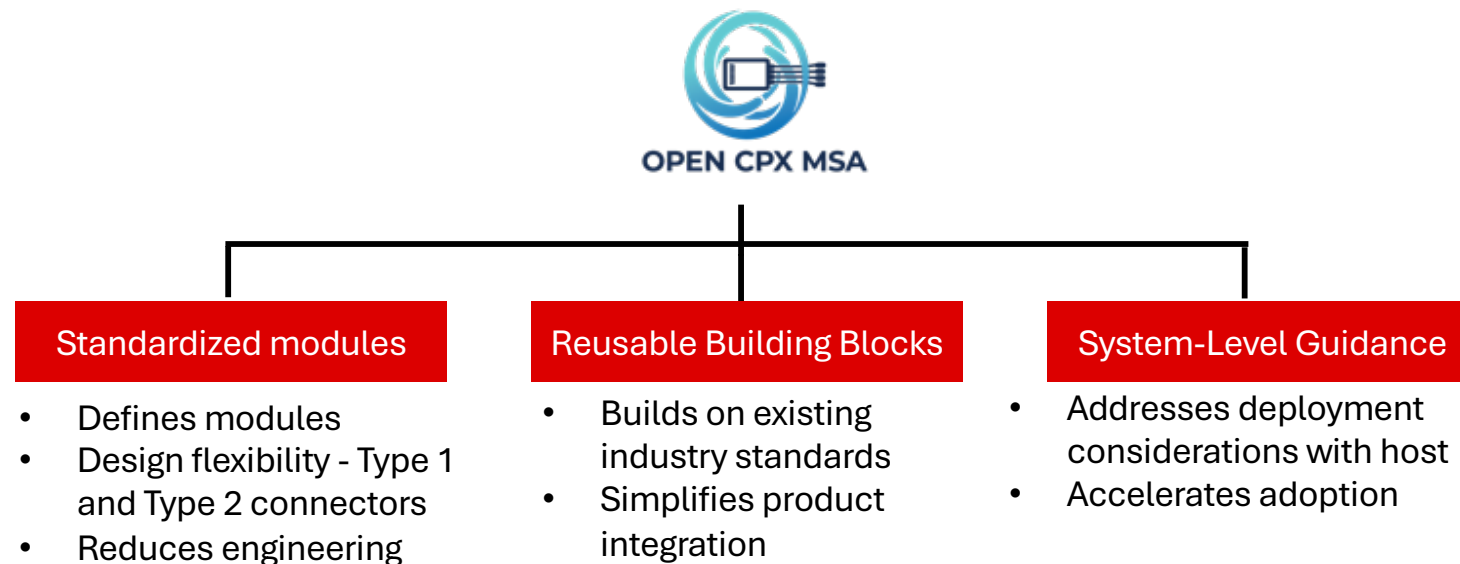


Open CPX MSA: Accelerating CPO/NPO Deployment



Open CPX MSA reduces time from innovation to deployable products.

Defines both modules and ecosystem to bring CPO/NPO products to market faster.



Version 1 Specification
Available Q3 2026
***Less than 8 months
from project launch***

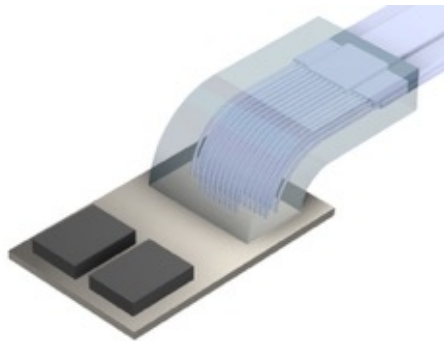
Accelerating the path from innovation to deployable CPO/NPO products.

Example of Open CPX-compliant optical engine



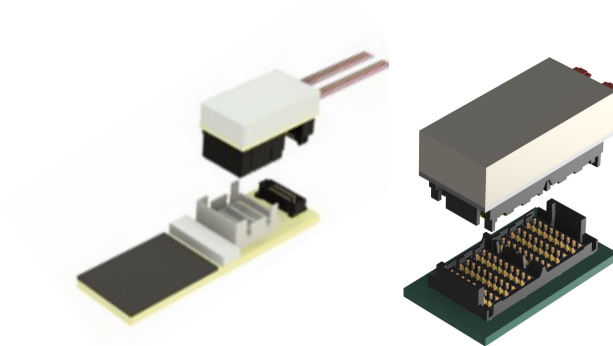
High density, low power pluggable CPO/NPO

- 6.4T within CPX connector
- ASICs up to 200T
- Up to 70% power reduction versus retimed



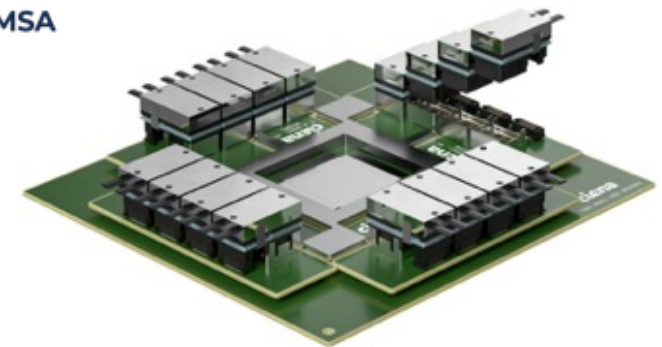
High Reliability

- Proven PIC and EIC technologies
- Field replaceable ELSFP
- Press on connector, Type 1 or Type 2



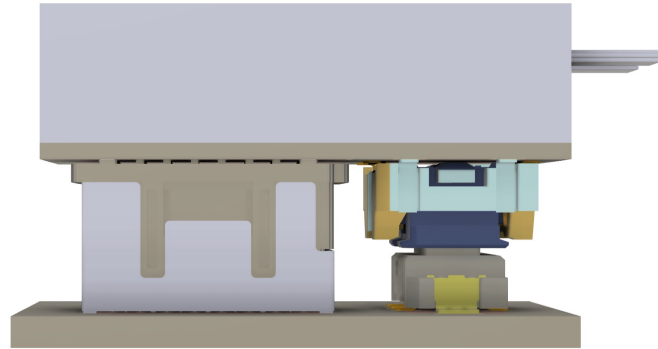
Paves way for open ecosystem CPO/NPO

- Disaggregated
- Standards based

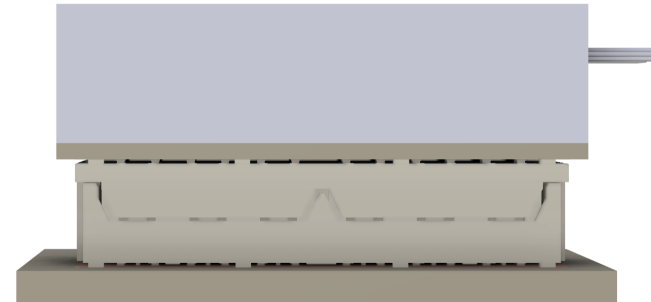


Removing the obstacles to widespread NPO/CPO adoption

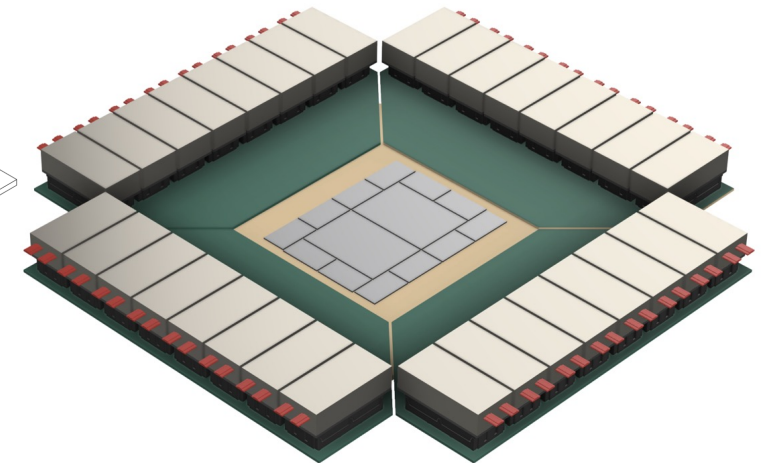
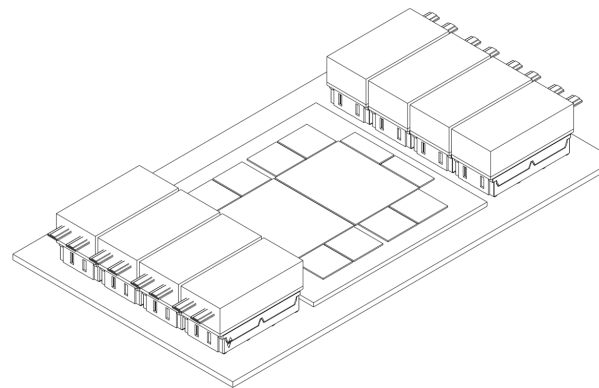
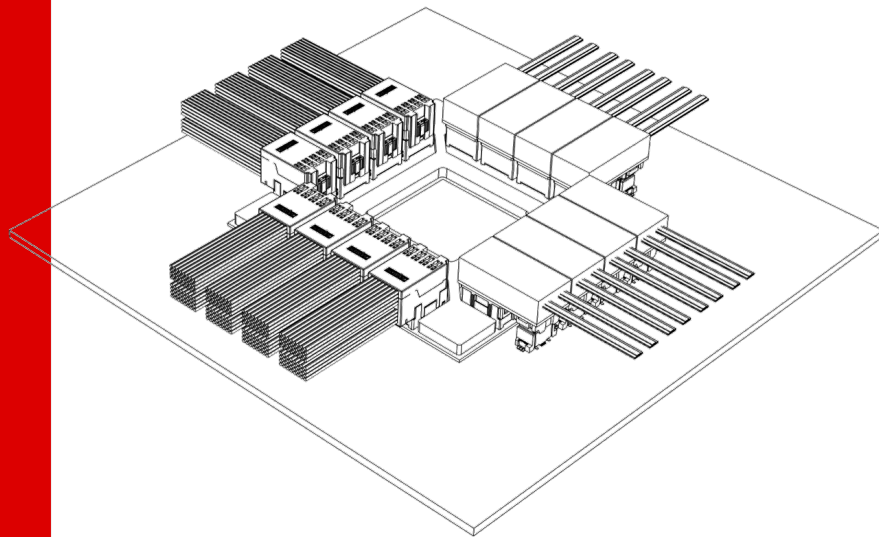
OpenCPX MSA Module Types



Type 1

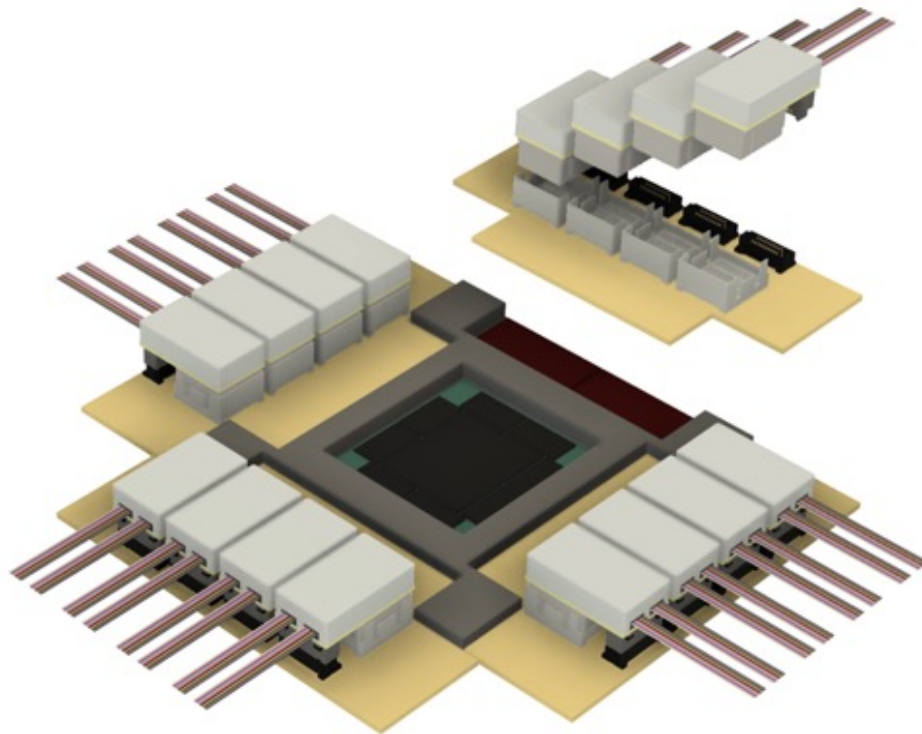


Type 2

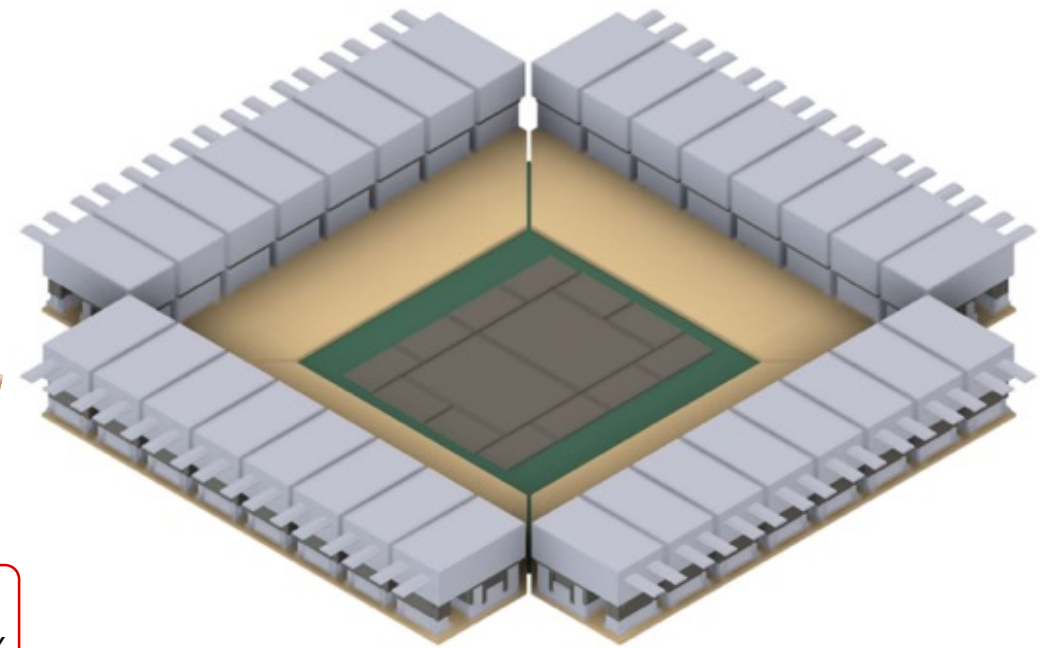


Support for 102.4T with a path to 204.8T using 6.4T IO

6.4T CPO configuration
supporting 102.4T Switch ASIC



6.4T CPO configuration
supporting 204.8T Switch ASIC



ciena
Vesta 200 6.4T CPX
Optical engine



External lasers enable a narrower optical engine footprint, providing a clear and scalable path from 102.4T to 204.8T platforms.

Technology choices expanding for data center networks

At higher speeds, Coherent-Lite needed for campus and even scale-out applications

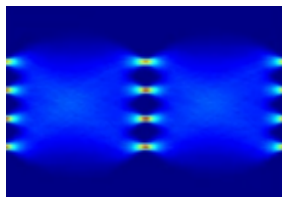
Electrons

Photons



DAC - Direct Attach Copper,
ACC - Active Copper Cable

IMDD Electrical



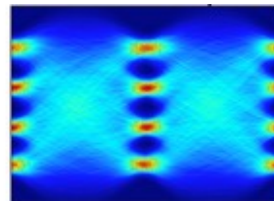
10's m $\rightarrow$ 1 m

Most suitable for short
or lower-speed interconnect
within a rack



IMDD DSP, xPO

IMDD Optical

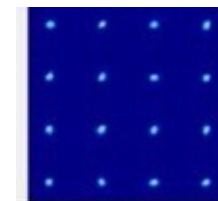


2 km $\rightarrow$ 10's m

Runs out of gas
at higher speed and distance



Coherent-Lite

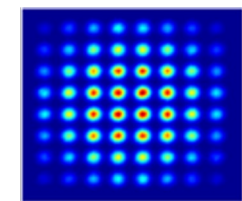


20 $\rightarrow$ 2 km

Will be required for 1.6T/3.2T+
and for longer distance interconnect



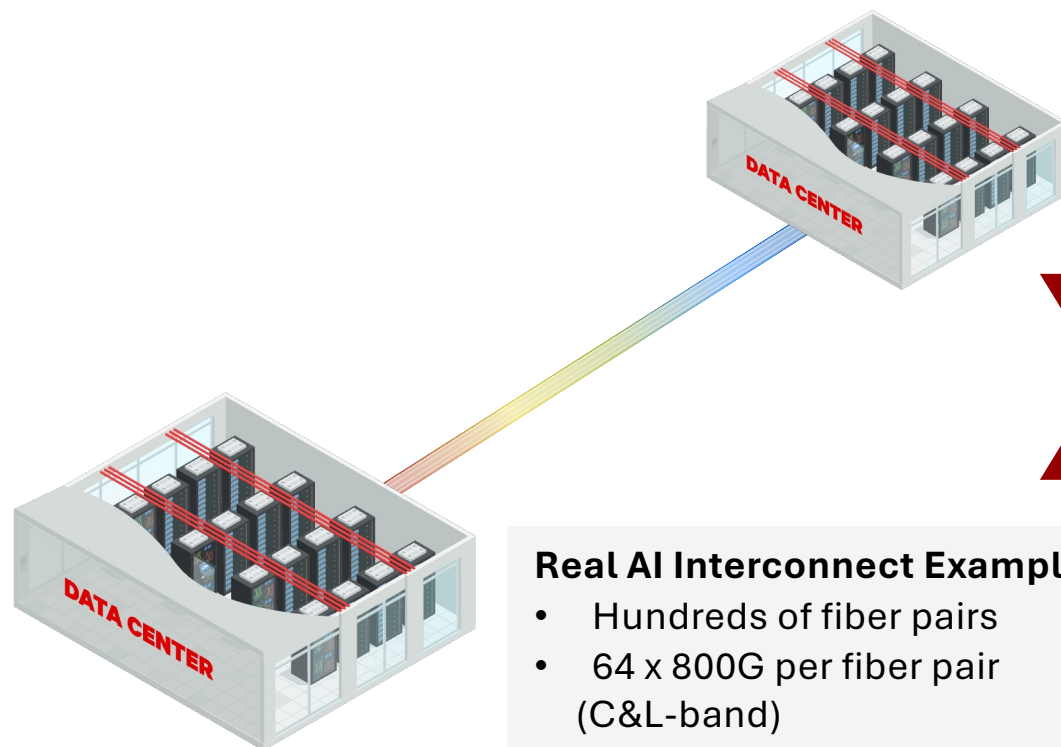
Coherent



100 $\rightarrow$ 20 km

The rise of Scale-Across networking is here

SCALE-ACROSS SOLUTION



Real AI Interconnect Example

- Hundreds of fiber pairs
- 64 x 800G per fiber pair (C&L-band)

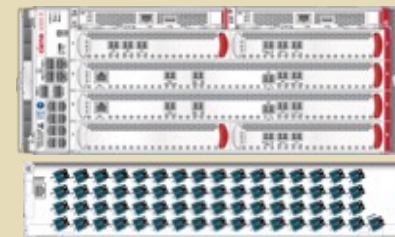
WaveLogic coherent optics



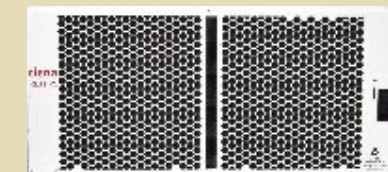
- WaveLogic 6 Nano 800G ZR+ with interoperable PCS
- C- and L-Band variants

RLS modern line system

RLS C&L terminal config

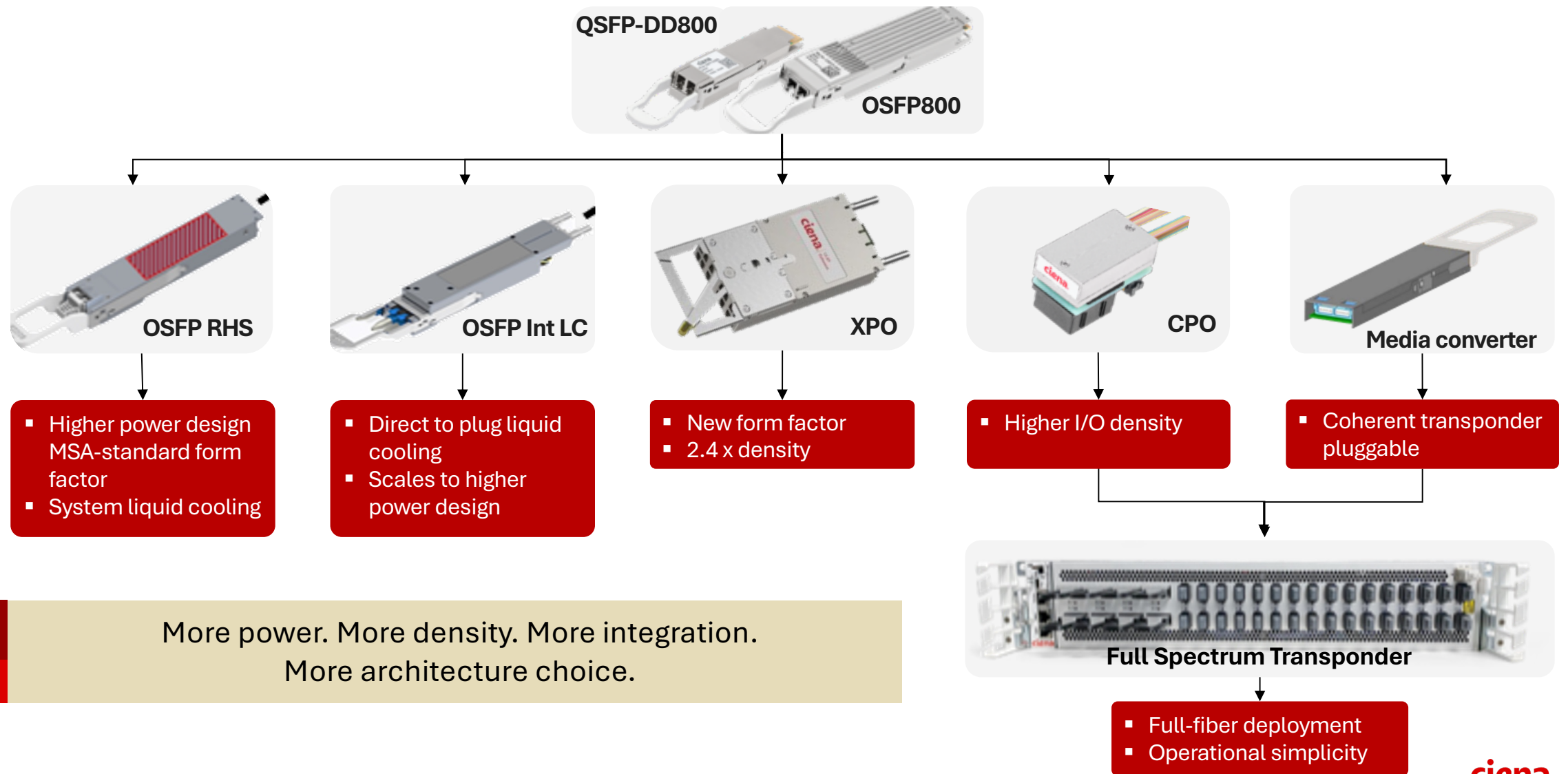


RLS Hyper-Rail



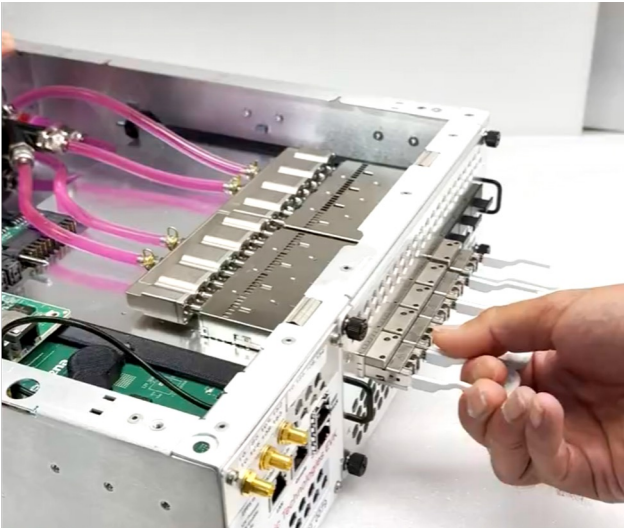
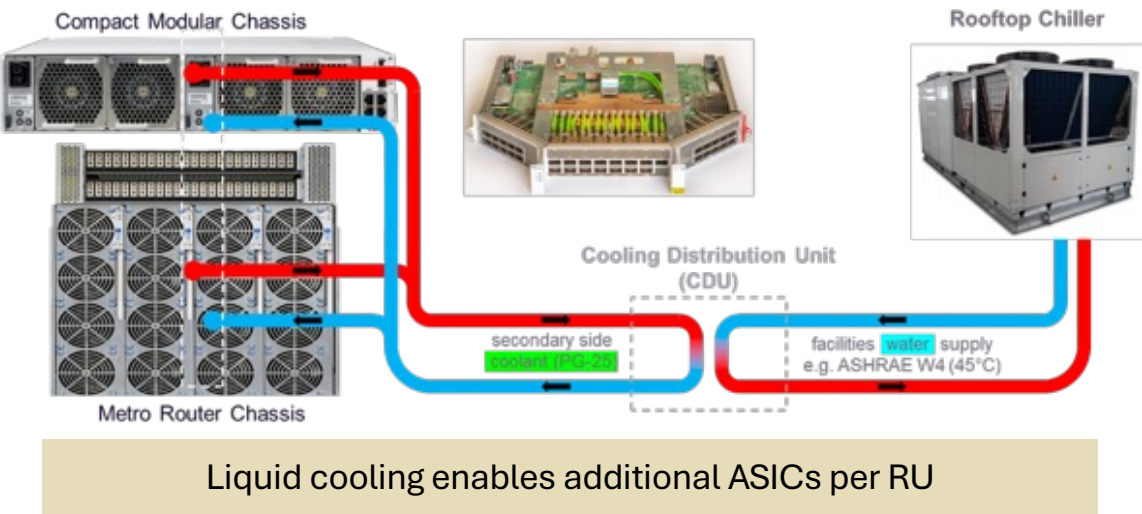
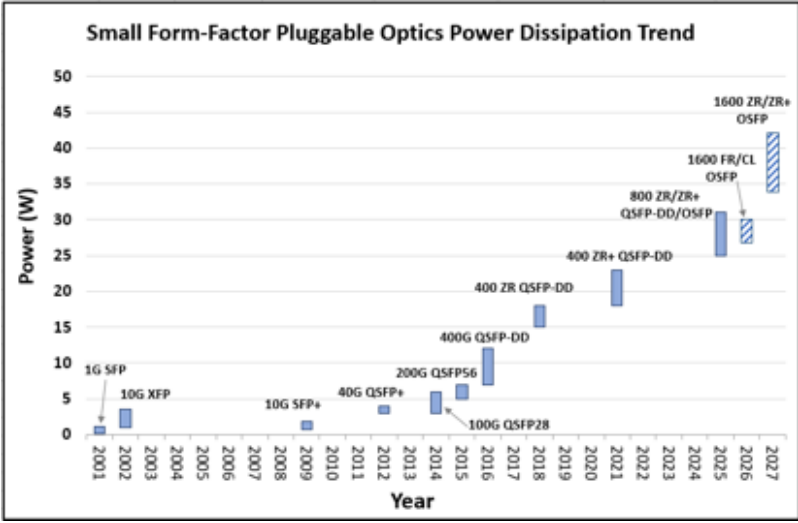
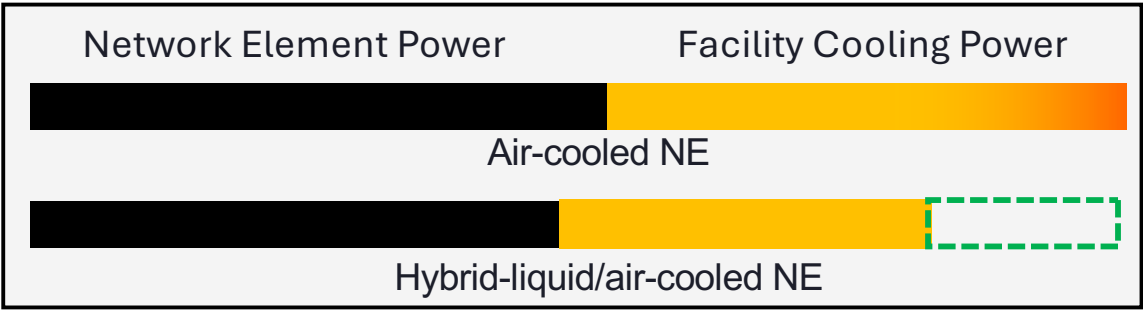
16 x C- and L-Band fiber pairs in 5RU

Form factors evolving to meet new system requirements



Liquid cooling inevitable with substantial advantages

- OPEX savings thru facility energy consumption, water consumption, GHG emissions
- Eliminates fan power consumption and system fan noise.

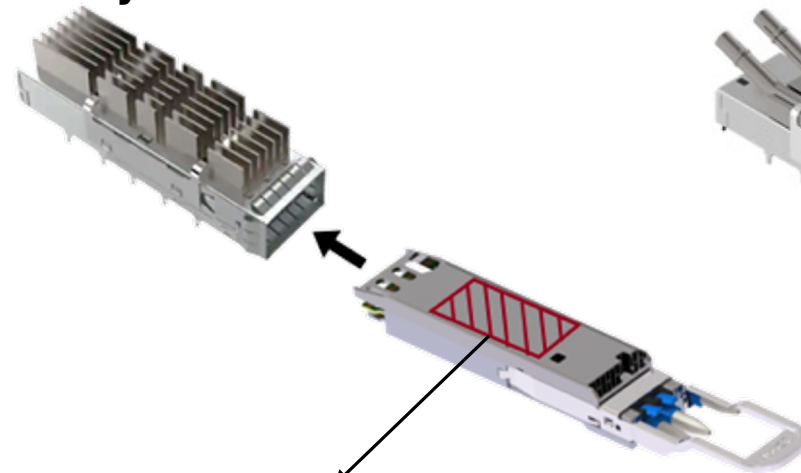


12.8T XPO

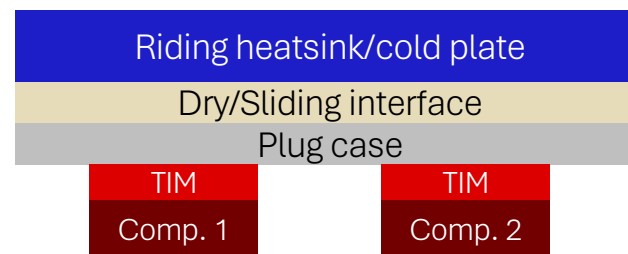
Direct-to-plug cooling enables higher-power pluggables with fewer thermal constraints

Riding Air-Cooled Solution

Today's solution

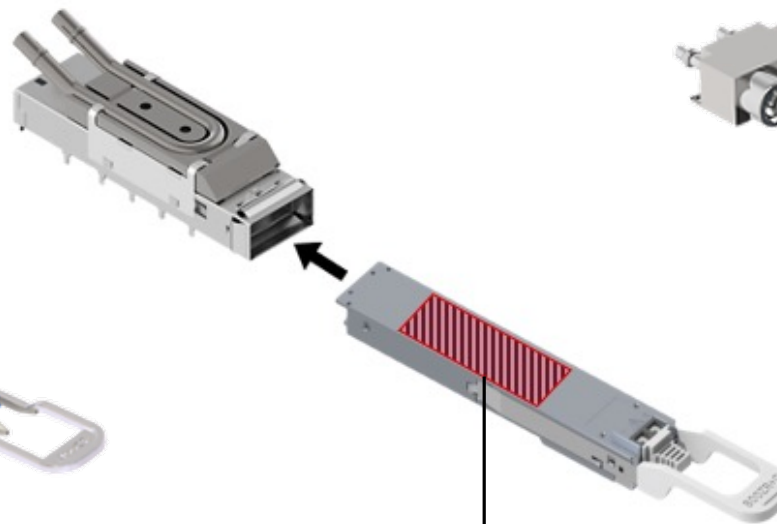


x Thermal Penalty



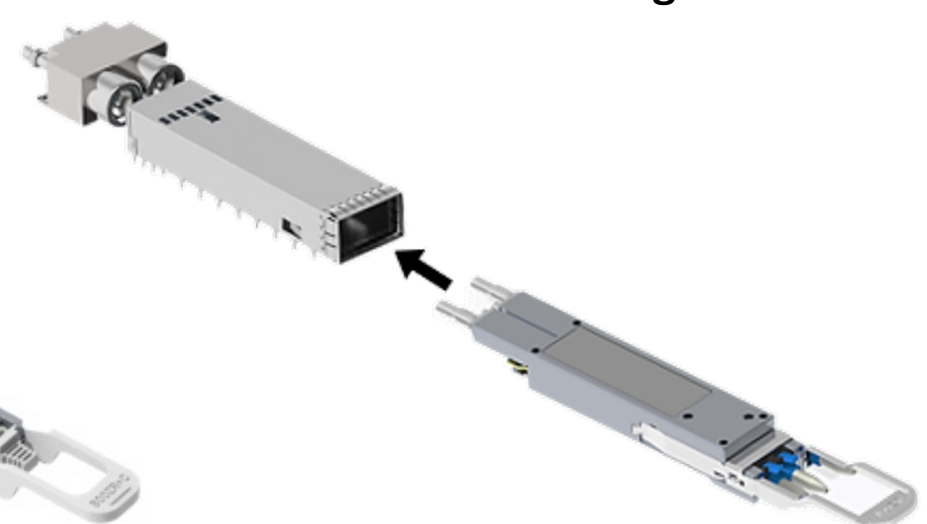
Riding Liquid-Cooled Solution

Existing OSFP MSA-compliant



Direct-To-Plug Liquid-Cooled Solution

New mechanical design



- ✓ No external heatsink/cold plate
- ✓ No dry/sliding contact



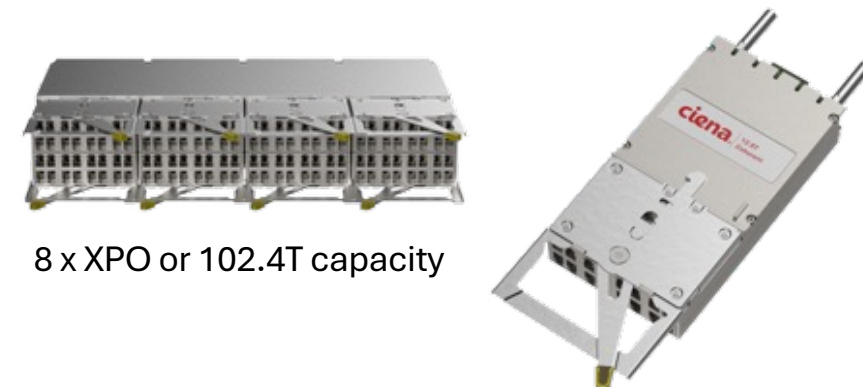
XPO — scaling density with front-panel pluggability

Scaling routers and pluggables

- Standard, flexible liquid-cooled 12.8T pluggable
- Higher density routers – 200T in 1RU
- Robust ecosystem with XPO MSA
- Flexible thermal envelope enables products for: Scale Up, Scale Out, Scale Across and beyond
- Liquid Cooling provides important design and reliability benefits

XPO broadens the architectural choices available to customers, enabling higher density while preserving front-panel pluggability.

XPO
OPTICS INNOVATION



8 x XPO or 102.4T capacity



12.8T Linear Drive

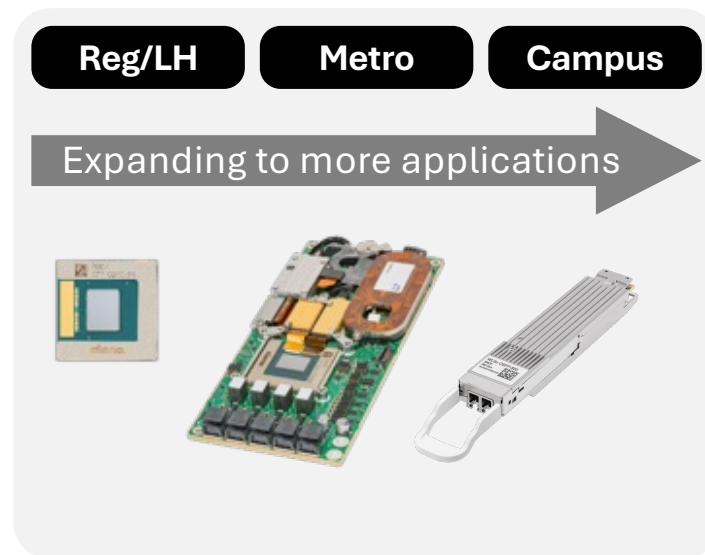
12.8T
Coherent-Lite

12.8T Coherent ZR

ciena

Optical interconnect evolution

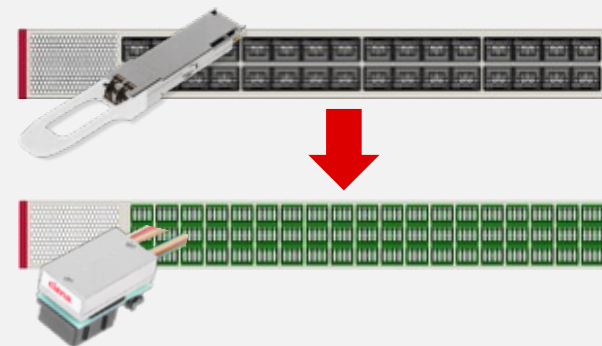
Coherent evolution



IMDD evolution

Co-packaged optics (CPO)

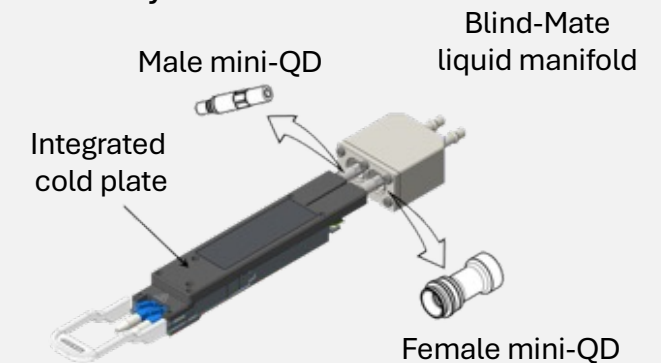
- Higher density
- Lower electrical power



Liquid cooling

Direct-to-plug liquid cooling

- Enables higher-power pluggables and systems



AI requirements are causing coherent, IMDD, and thermal innovation to converge into entirely new classes of optical systems

12.8T XPO (8 x 1.6T)

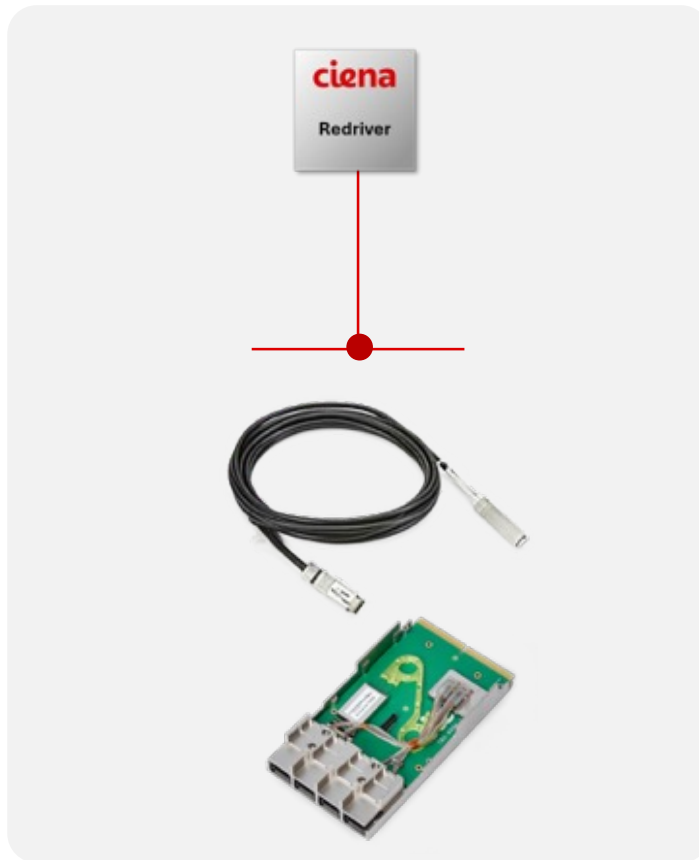


51.2T Full Spectrum Transponder, with CPO clients

Examples of interconnect technology implementation

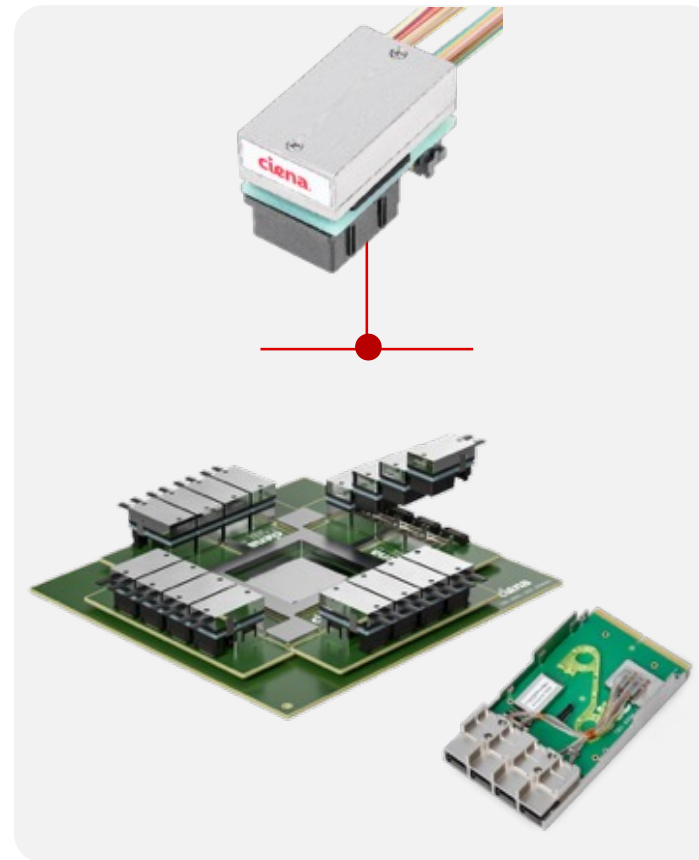
Scale up

Nitro 200



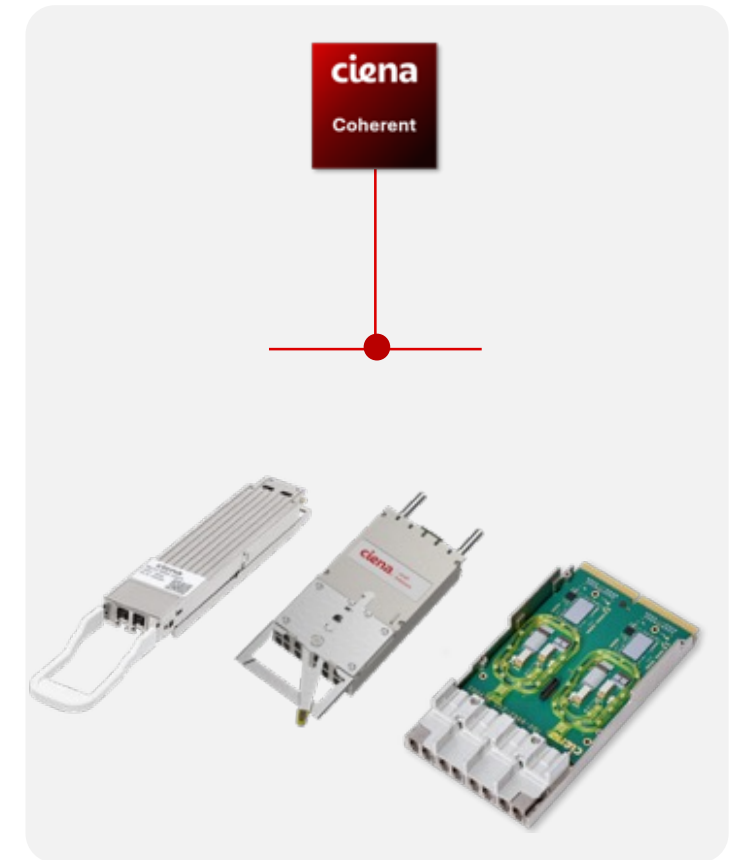
Scale out

Vesta 200

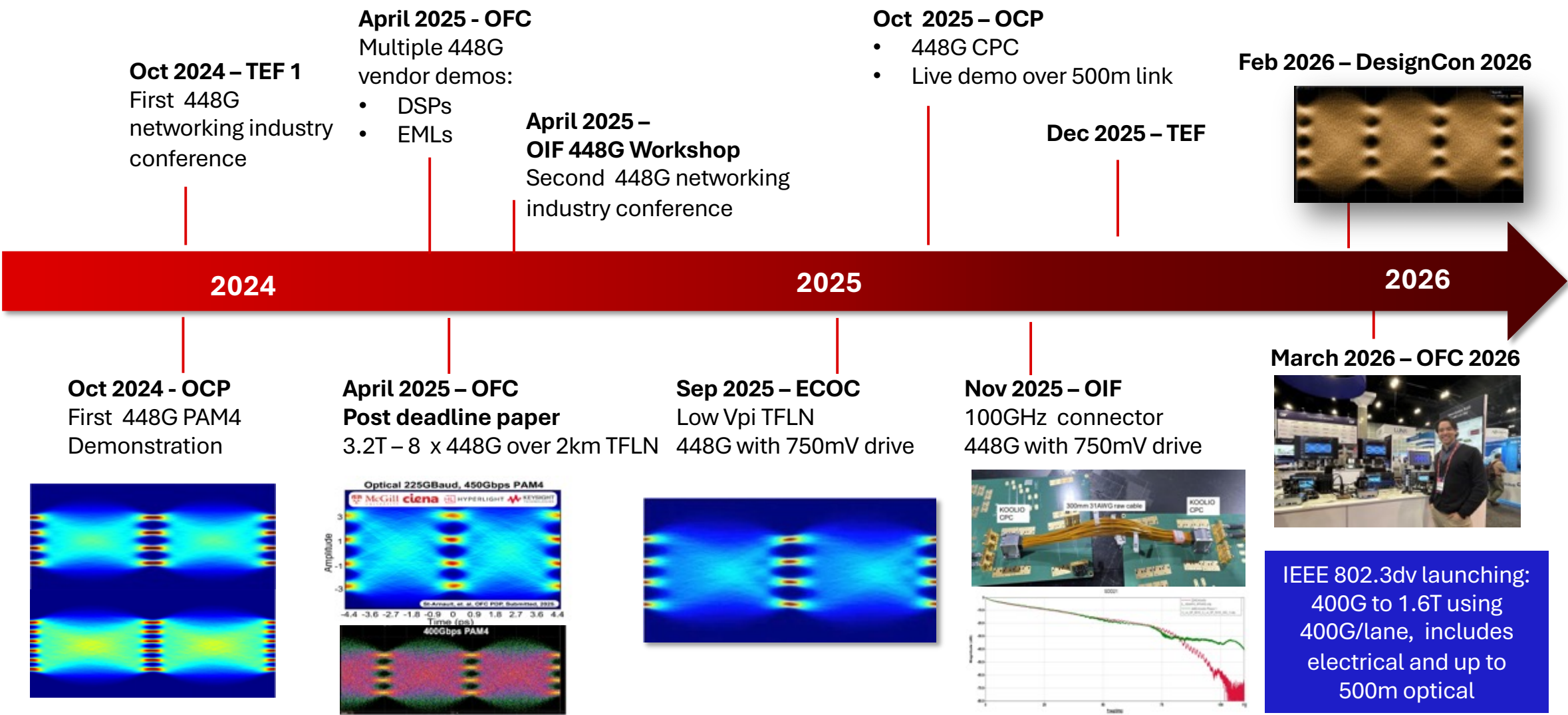


Campus / Scale across

WaveLogic Nano



Looking ahead - 448G/lane ecosystem progress



Takeaways

- 1 AI is blurring the boundaries between scale up, scale out, and scale across.
- 2 The next scaling challenge is not just bandwidth — it is bandwidth density, power, and reach.
- 3 Faster SerDes and closer-to-compute optics are complementary levers for continued scaling.
- 4 No single technology wins everywhere: copper, IMDD, coherent-lite, and coherent each have a role.
- 5 Open ecosystems and new thermal architectures will be essential to deploying these technologies at AI scale.

The future of AI interconnects will be defined less by a single winning technology than by how effectively we combine the right technologies across distance, power, density, and scale.

Thank You